Competitor Benchmark Study

Suspension and ride systems - leaf spring and emerging composite spring technologies, comparable-tonnage platform

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Scope	Front and rear leaf spring systems, comparable tonnage
Benchmarks examined	4 platforms
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Summary

Leaf spring systems across four comparable-tonnage platforms were examined for mass, rate characteristic and cost. The principal finding concerns a glass-fibre-reinforced composite mono-leaf spring observed on one benchmark, reported at a 35 per cent mass saving against an equivalent steel multi-leaf design, with durability validation still ongoing at the time of that report.

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1 Scope and method

This study examines leaf spring systems on four comparable-tonnage platforms, covering front and rear suspensions. Springs were removed, weighed, measured for free camber and rate-tested on a static rig. Where a platform used a construction not currently in use on the subject vehicle, the construction is described in detail rather than only compared on mass.

1.1 Rate testing

Each spring was loaded to its rated capacity in ten increments and unloaded in the same increments, with deflection recorded at each step. Loading and unloading curves are reported separately because the area between them, the hysteresis arising from interleaf friction, is a material difference between constructions and is not visible from a single rate figure.

1.2 Constructions examined

- Steel multi-leaf, nine leaves, the construction in use on the subject vehicle rear suspension.
- Steel parabolic, three leaves, in use on the subject vehicle front suspension and on two benchmarks.
- Steel parabolic, two leaves, observed on one benchmark with a higher-rate interleaf pad.
- GFRP composite mono-leaf, observed on one benchmark, reported in section 2.1.

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1.3 Measured masses and rates

Construction	Mass per side (kg)	Rate (N/mm)	Hysteresis (% of rate)
Steel multi-leaf, 9 leaf	86.4	428	11.2
Steel parabolic, 3 leaf	62.8	416	6.4
Steel parabolic, 2 leaf	54.1	421	5.1
GFRP composite mono-leaf	36.5	409	1.8

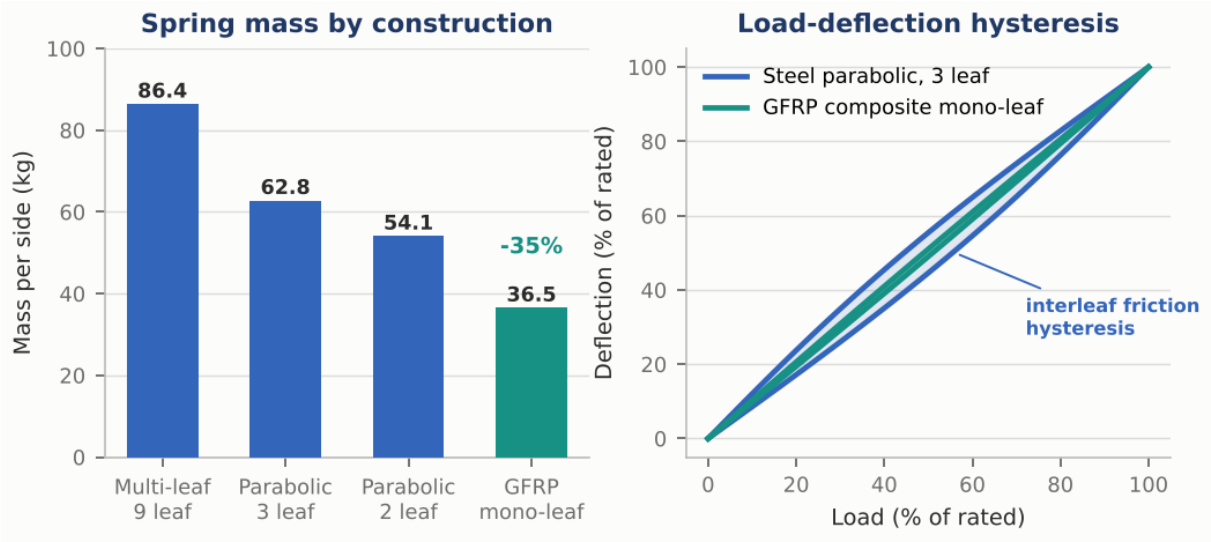


Figure 1 - Spring mass by construction, and load-deflection hysteresis. The area between the loading and unloading curves is interleaf friction, which a mono-leaf construction largely removes.

Rate is comparable across all four constructions, as it must be for a like-for-like ride comparison. Mass and hysteresis are not.

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1.4 Cost comparison

Construction	Material	Process	Est. cost per side (INR)
Steel multi-leaf, 9 leaf	55Si7 spring steel	Hot form, quench, shot peen	6,840
Steel parabolic, 3 leaf	55Si7 spring steel	Taper roll, quench, shot peen	7,920
Steel parabolic, 2 leaf	55Si7 spring steel	Taper roll, quench, shot peen	7,140
GFRP composite mono-leaf	E-glass / epoxy	Filament wind and compression mould	18,600

Finding 1.1. The composite spring is 2.3 times the cost of the parabolic steel spring it would replace. At current volumes it does not meet the committee benefit threshold on cost alone; its case rests on mass and on the secondary benefits described in section 2.1.

1.5 Durability testing observed

Two of the four benchmarks published durability data for their spring systems. Neither published data for the composite construction, which was described in the source material as being in validation rather than in series production.

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2 Suspension sub-system findings

Three findings arise from the suspension comparison. The first two concern steel constructions already within the platform's capability; the third, reported in section 2.1, concerns a construction not currently in use.

2.0.1 Two-leaf parabolic front spring

One benchmark achieves a two-leaf front parabolic spring at the same rate as the subject vehicle's three-leaf spring, using a higher-rate interleaf pad to control the transition. The saving is 8.7 kg per side. Ride behaviour at part load was not assessed in this study and is the open question.

2.0.2 Hysteresis and ride

Hysteresis falls from 11.2 per cent of rate on the nine-leaf construction to 5.1 per cent on the two-leaf parabolic. Lower hysteresis improves small-input ride but reduces inherent damping, which must then be recovered by the shock absorber. A change of construction is therefore a suspension system change, not a spring change, and should not be proposed as a component substitution.

> Finding 2.1. A spring construction change alters the damping balance of the suspension. Proposals that treat it as a like-for-like component substitution are incomplete and should be returned for a system-level assessment.

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2.1 Composite mono-leaf spring

One benchmark platform fits a glass-fibre-reinforced polymer composite mono-leaf spring on the front axle. The construction is a single tapered E-glass and epoxy leaf with bonded steel end fittings and a moulded centre clamp seat.

2.1.1 Measured performance

Attribute	GFRP mono-leaf	Steel parabolic, 3 leaf	Difference
Mass per side (kg)	36.5	56.2	-35.1%
Rate (N/mm)	409	416	-1.7%
Hysteresis (% of rate)	1.8	6.4	-72%
Free camber (mm)	118	124	-4.8%
Est. cost per side (INR)	18,600	7,920	+135%

Finding 2.2. A glass-fibre-reinforced composite mono-leaf spring on a comparable-tonnage platform is reported with a 35 per cent mass saving against the steel multi-leaf design. Durability validation was still ongoing at the time of the source report.

2.1.2 Open questions

- Durability. No published fatigue data. Composite springs fail by delamination rather than by crack propagation, so steel fatigue criteria do not transfer.
- End fitting. The bonded steel fitting is the critical interface and could not be examined without destroying the sample.
- Stone impact. The front spring is directly exposed. Impact damage tolerance of the composite is unknown.
- Damping. Hysteresis falls by 72 per cent, which must be recovered in the shock absorber, as discussed in finding 2.1.
- Cost. 2.3 times the steel spring at current volumes.

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3 Recommendations

- Open a study on the two-leaf parabolic front spring, with a ride assessment at part load as the first gate. Mass benefit 8.7 kg per side.
- Record finding 2.1 in the idea register so that spring construction proposals are assessed as suspension system changes rather than component substitutions.
- Do not open a composite spring programme on this evidence. Revisit when durability data is published or when a supplier will share validation results under agreement.
- Re-test the composite case against material rates at each half-yearly cost review; its viability is volume and resin-price sensitive.

Status. The composite spring is recorded as a watch item, not a rejected idea. Proposals referencing it should be linked to this study rather than assessed from the beginning.

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